

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

**COURSE NAME: ARTIFICIAL
INTELLIGENCE AND EXPERT
SYSTEMS**

**COURSE OUTCOME COVERED
COURSE CODE: MLA01**

CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)

Assessment Tool Weightage: 30%

Total Marks: 100 Time: 90 Mins No. of Questions: 1

Annexure A

Research Review & Analysis

Code of Conduct:

I Namburi Sai Yashaswini (192425250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: yashaswini

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	

Total	100%	
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Signature of the Faculty Total Marks Awarded

Decision Tree Learning: A Comparative Review of Attribute Selection, Pruning, and Performance Characteristics in ID3, C4.5, and CART

Abstract

Decision trees remain among the most widely used classification and regression models in artificial intelligence and data mining because their hierarchical, rule-like structure is easy for humans to interpret. Three algorithms — ID3, C4.5, and CART — form the historical and technical foundation of nearly all modern tree-based learners, including ensemble methods such as Random Forests and Gradient Boosted Trees. This review examines and compares these three algorithms with respect to their attribute-selection criteria (information gain, gain ratio, and Gini impurity / least-squares deviation), their pruning strategies (none, post-pruning, and cost-complexity pruning), their ability to handle continuous and missing values, their interpretability, their computational complexity, and their classification performance as reported across the reviewed literature. The methodology followed is a structured narrative literature review of classical foundational papers together with more recent comparative and survey studies. The review finds that while ID3 remains conceptually important, its inability to handle continuous attributes or control overfitting limits its practical use; C4.5 and CART address these limitations in different ways, with CART's strict binary splitting and cross validated cost-complexity pruning giving it an advantage in resistance to overfitting, while C4.5's gain ratio criterion reduces bias toward high-cardinality attributes. A recurring research gap identified across the reviewed studies is the absence of a unified, computationally efficient criterion that simultaneously optimises accuracy, interpretability, and robustness to noisy or imbalanced data without requiring extensive parameter tuning. Building on this gap, the review proposes a hybrid, validation-aware attribute-selection and pruning framework that adapts its splitting criterion to attribute type and dataset characteristics at each node, combined with an adaptive, statistically grounded stopping rule. The review concludes that continued hybridisation of classical decision tree strategies, rather than complete replacement by opaque

models, offers a promising direction for building AI systems that are simultaneously accurate and explainable.

Keywords

Decision Tree Learning; ID3 Algorithm; C4.5 Algorithm; CART; Attribute Selection; Pruning Strategies; Interpretable Machine Learning

1. Introduction

Decision tree learning is one of the earliest and most enduring approaches to inductive supervised learning in artificial intelligence. A decision tree represents a hypothesis about a target concept as a hierarchy of tests on input attributes, where internal nodes correspond to attribute tests, branches correspond to test outcomes, and leaf nodes correspond to predicted class labels or numeric outputs. This structure mirrors the way humans often reason through a sequence of yes/no or threshold-based questions, which is why decision trees are frequently described as “white-box” or self-explanatory models, in contrast to “black-box” models such as deep neural networks.

The importance of decision tree learning in AI and data mining stems from three complementary strengths: (i) the resulting models are directly interpretable by domain experts, which is critical in regulated or high-stakes domains such as healthcare, finance, and law; (ii) the training procedure is computationally efficient relative to many other learning paradigms, making decision trees suitable for large tabular datasets; and (iii) decision trees form the building blocks of powerful ensemble methods, including bagging-based Random Forests and boosting-based Gradient Boosted Decision Trees, both of which remain state of the art on many structured/tabular prediction tasks even in the era of deep learning.

Three algorithms anchor the field historically and pedagogically. ID3 (Iterative Dichotomiser 3), introduced by Quinlan, builds a tree top-down using information gain derived from Shannon entropy to select the attribute that best separates the training examples at each node. C4.5, also developed by Quinlan as a direct successor to ID3, extends the approach with a gain-ratio criterion that corrects information gain's bias toward attributes with many values, native handling of continuous attributes through threshold discretisation, handling of missing values, and, crucially, a post-pruning mechanism that improves generalisation. CART (Classification and Regression Trees), introduced by Breiman, Friedman, Olshen, and Stone, generalises the paradigm further by supporting both classification and regression through Gini impurity and least-squares deviation respectively, enforcing strictly binary splits, and applying cost complexity pruning validated through cross-validation.

Despite the maturity of these algorithms, they continue to be actively revisited in current research, both as standalone interpretable models and as base learners inside ensembles, automated machine learning (AutoML) pipelines, and explainable-AI (XAI) frameworks. Understanding precisely how ID3, C4.5, and CART differ in their attribute-selection logic, their treatment of overfitting through pruning, their computational cost, and their empirical accuracy is therefore still practically important for anyone selecting or designing a tree-based learning system.

This review is scoped to a comparative analysis of the three algorithms as applied to supervised classification (and, for CART, regression) problems on structured/tabular data. Deep architectural variants (e.g., oblique trees, fuzzy decision trees) and ensemble methods are referenced only where they clarify how the base algorithms have been extended, since a full review of ensemble learning is outside the present scope.

2. Research Problem and Objectives

Although ID3, C4.5, and CART are frequently mentioned together in textbooks and survey papers as “classical decision tree algorithms,” many comparative treatments describe them at a surface level — for example, simply stating that C4.5 “improves on” ID3 — without systematically decomposing the specific mechanisms responsible for each algorithm's behaviour on accuracy, interpretability, and computational cost. This creates a practical research problem: practitioners and students often cannot clearly determine which algorithm is preferable for a given dataset characteristic (e.g., presence of continuous attributes, high cardinality categorical attributes, missing values, or class imbalance), because the trade-offs are rarely presented in a single, structured comparative framework grounded in both the original algorithmic specifications and subsequent empirical studies.

Objectives of the Review

- 1.** To review and summarise the attribute-selection mechanisms of ID3 (information gain), C4.5 (gain ratio), and CART (Gini impurity / least-squares deviation), including their mathematical basis.
- 2.** To compare the pruning strategies used by each algorithm — no pruning (ID3), post pruning (C4.5), and cost-complexity pruning with cross-validation (CART) — and analyse their effect on overfitting and generalisation.
- 3.** To evaluate each algorithm along the dimensions of interpretability, computational complexity, ability to handle continuous and missing values, and reported classification performance.

4. To identify a concrete research gap in the existing comparative literature on decision tree algorithms.
5. To propose a feasible improvement or future research direction that addresses the identified gap.

3. Literature Review

3.1 ID3 and the Information-Gain Paradigm

ID3 was formalised by Quinlan as an implementation of Hunt's concept-learning framework, in which a tree is grown recursively by choosing, at each node, the attribute that yields the greatest reduction in entropy (uncertainty) among the training examples. Foundational treatments of ID3 (e.g., Quinlan's original work and subsequent textbook expositions such as Rokach and Maimon's decision-tree chapter) emphasise that ID3 handles only discrete/categorical attributes, has no mechanism for missing values, and produces a fully grown tree with no built-in pruning step — a design choice that later research consistently identifies as ID3's principal weakness because it leaves the algorithm prone to overfitting on noisy training data.

Comparative surveys such as Hssina et al.'s widely cited study of ID3 and C4.5 note that ID3's information-gain criterion is systematically biased toward attributes with a large number of distinct values (e.g., an identifier-like column), because splitting on such attributes trivially reduces entropy without necessarily improving generalisation. This observation motivated the gain-ratio correction subsequently introduced in C4.5.

3.2 C4.5: Gain Ratio, Continuous Attributes, and Post-Pruning

C4.5, also due to Quinlan, was developed explicitly to correct ID3's shortcomings. The literature reviewed here consistently highlights three extensions: (i) a gain-ratio criterion that normalises information gain by the intrinsic information (split information) of the candidate attribute, discouraging splits on high-cardinality attributes; (ii) native support for continuous attributes by dynamically searching for the best binary threshold; and (iii) handling of missing attribute values by probabilistically distributing fractional instances across branches. Reviews such as the comparative survey by researchers examining ID3, CART, and C4.5 together describe C4.5's post-pruning — typically pessimistic error-based pruning or subtree replacement/raising — as a mechanism that intentionally increases the error rate on training data in exchange for a measurable decrease in error on unseen test data, i.e., improved generalisation.

3.3 CART: Binary Splits, Gini Impurity, and Cost-Complexity Pruning

CART, introduced by Breiman, Friedman, Olshen, and Stone, differs structurally from both ID3 and C4.5 in that it always produces a strictly binary tree regardless of whether the underlying attribute is categorical or continuous, and it supports regression as well as classification through the least-squares-deviation criterion in addition to the Gini impurity criterion used for classification. Reviewed comparative papers (e.g., the Springer-published performance evaluation study of ID3, C4.5, and CART, and related survey work) report that CART's cost-complexity pruning — in which a maximal tree is deliberately overgrown and then trimmed back using a complexity parameter selected via cross-validation — tends to produce more compact, stable trees that are less sensitive to small perturbations of the training set than unpruned ID3 trees.

3.4 Cross-Cutting Comparative Studies

Several dedicated comparative papers evaluate all three algorithms together rather than in isolation. These studies generally agree on a small number of recurring findings: first, that C4.5 and CART tend to outperform plain ID3 in classification accuracy on datasets containing continuous or noisy attributes, precisely because of their pruning mechanisms; second, that CART's binary-split constraint can produce deeper trees than C4.5's multi-way splits on categorical data, which affects readability even though overall accuracy may be comparable; and third, that all three algorithms remain computationally efficient compared with iterative or gradient-based learners, since tree induction is fundamentally a greedy, single-pass recursive partitioning procedure. More recent expository work (including applied Python-based walkthroughs of the three splitting rules) continues to use the same three algorithms as the standard pedagogical reference point for introducing impurity-based learning, underscoring their continued relevance in current AI curricula and applied research.

4. Comparative Analysis

Table 1 synthesises the attribute-selection method, pruning strategy, data-handling capability, interpretability, computational complexity, and typical reported performance of ID3, C4.5, and CART, drawing on the sources discussed in Section 3.

Attribute-selection measure	Information Gain (entropy reduction)	Gain Ratio (Information Gain / Split Information)	Gini Impurity (classification) / Least-Squares Deviation

			(regression)
Tree structure	Multi-way splits (one branch per attribute value)	Multi-way for categorical; binary threshold for continuous	Strictly binary splits for all attribute types
Continuous attributes	Not supported natively (requires pre discretisation)	Supported via dynamic threshold search	Supported via binary threshold search
Missing values	Not handled	Handled via probabilistic fractional instances	Handled via surrogate splits
Pruning strategy	None (fully grown tree)	Post-pruning (pessimistic error based subtree replacement/raising)	Cost-complexity pruning validated by cross-validation
Bias toward high cardinality attributes	High (favours many valued attributes)	Reduced by gain-ratio normalisation	Reduced by binary split search, though can still favour granular attributes

Interpretability	High — shallow, rule like trees on clean categorical data	High — comparable to ID3, slightly deeper due to thresholds	Moderate — binary splits can increase depth versus multi way trees
Computational complexity	Lowest (no pruning, no threshold search)	Higher than ID3 (threshold search + pruning pass)	Comparable to or higher than C4.5 (exhaustive binary split search + cross validated pruning)

Robustness to noise / overfitting	Low — no mechanism to limit overfitting	Improved via post pruning	Strong — cross validated pruning explicitly targets generalisation
Typical use case	Small, clean, purely categorical datasets; teaching	General-purpose classification with mixed attribute types	Classification and regression; common base learner in bagging/boosting ensembles

Table 1. Comparative summary of ID3, C4.5, and CART across key parameters.

The qualitative comparison in Table 1 is illustrated visually in Figure 4, which synthesises the relative ratings implied by the reviewed literature across six practically important dimensions. The figure is intended as a synthesis aid rather than a report of a single controlled experiment, since, as discussed in Section 5, the reviewed studies do not use a common benchmark suite.

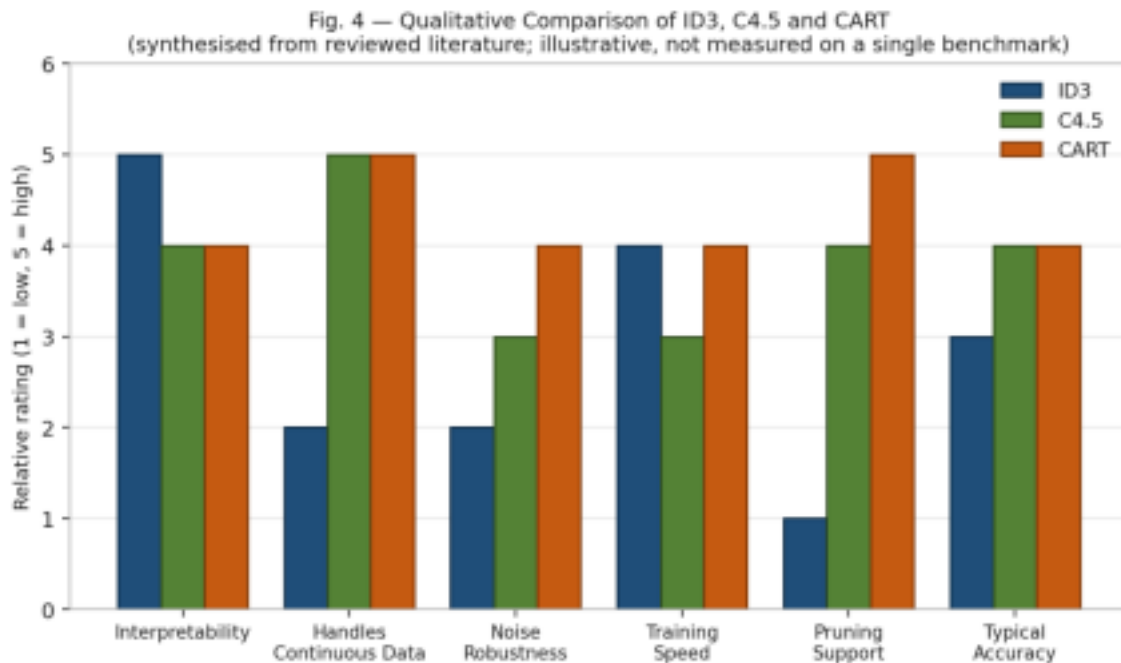


Fig. 4. Qualitative comparison of ID3, C4.5, and CART across six evaluation dimensions, synthesised from the reviewed literature.

Figures 1–3 below present the step-by-step control flow of each algorithm, drawn from the algorithmic descriptions in the reviewed literature, to make the differences in attribute selection and pruning logic concrete.

Fig. 1 — ID3 Algorithm: Attribute Selection Using Information Gain

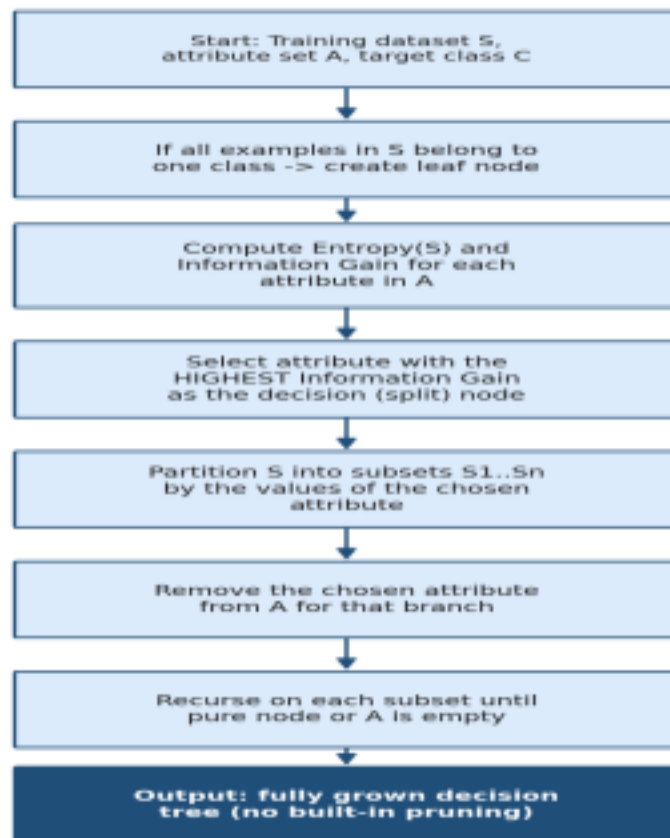


Fig. 1. ID3 algorithm — attribute selection using information gain, with no pruning step.

Fig. 2 — C4.5 Algorithm: Gain Ratio, Continuous Attributes and Post-Pruning

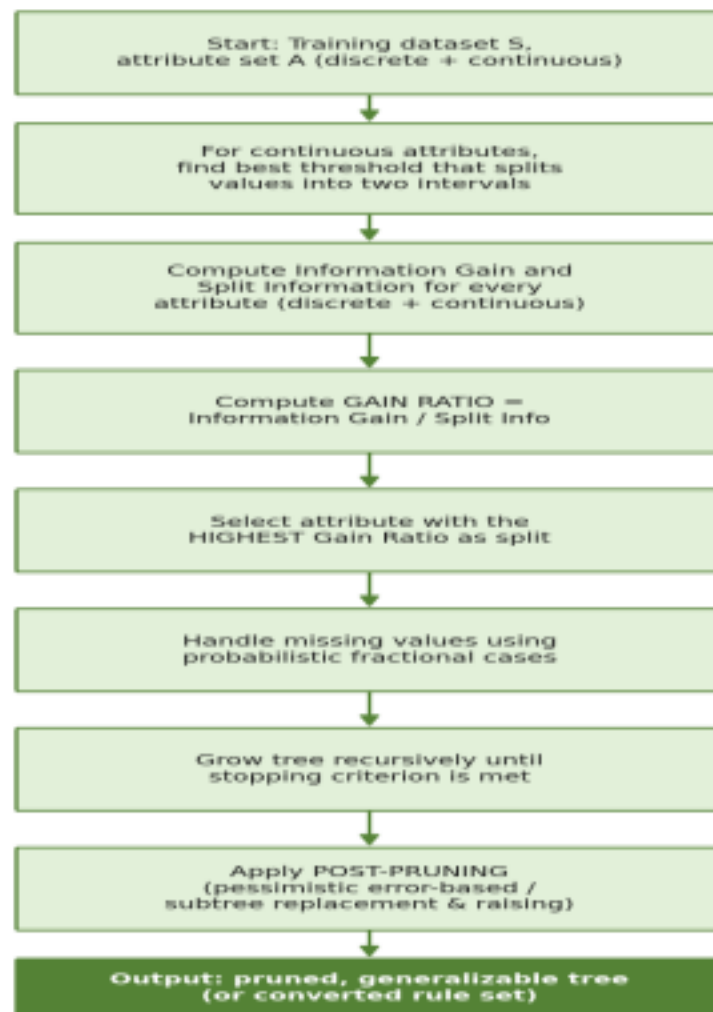


Fig. 2. C4.5 algorithm — gain ratio, continuous-attribute thresholding, and post-pruning.

Fig. 3 — CART Algorithm: Gini Index / LSD with Cost-Complexity Pruning

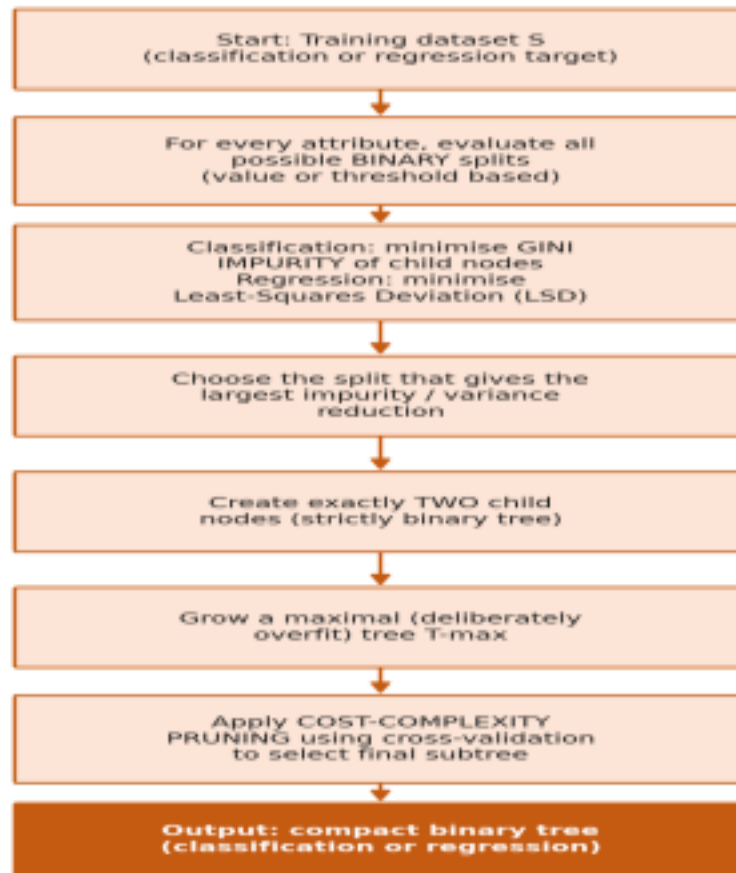
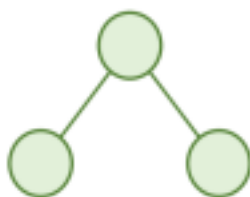


Fig. 3. CART algorithm — Gini impurity / least-squares deviation with binary splits and cost complexity pruning.

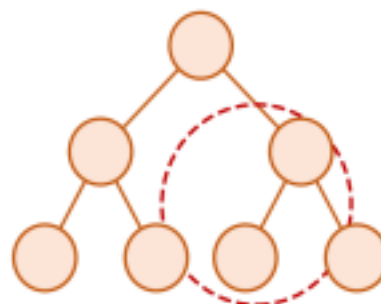
A further structural distinction discussed in the literature is the difference between pre-pruning (halting tree growth early) and post-pruning (growing a full tree and then trimming it back), illustrated schematically in Figure 5. ID3 uses neither; C4.5 uses post-pruning; CART uses a cross-validated form of post-pruning known as cost-complexity pruning.

Pre-Pruning (Early Stopping)



Growth halted early using a statistical test / min-samples threshold before full purity

Post-Pruning (Grow then Trim)



subtree pruned back (replaced by a leaf) after validation-error check

Fig. 5. Schematic contrast between pre-pruning (early stopping) and post-pruning (grow-then trim) strategies.

5. Critical Analysis

Across the reviewed literature, several strengths and weaknesses recur consistently. A clear strength shared by all three algorithms is computational tractability: because tree induction proceeds by a greedy, top-down, recursive partitioning strategy rather than an iterative optimisation over a continuous parameter space, all three algorithms train quickly even on moderately large tabular datasets, which explains their continued use as base learners inside modern ensemble methods.

A shared weakness, however, is that all three algorithms are greedy: the locally optimal attribute chosen at a given node is never reconsidered later, so none of the three algorithms guarantees a globally optimal tree with respect to any of the underlying impurity measures. The reviewed comparative studies rarely discuss this limitation explicitly, instead focusing on empirical accuracy differences, which is itself a limitation of the existing literature (see Section 6).

ID3's principal weakness — the complete absence of pruning — is well documented and is the most consistently cited reason for its inferior performance relative to C4.5 and CART on datasets that contain noise or continuous attributes forced into categorical form. At the same time, ID3's simplicity remains pedagogically valuable, and several reviewed sources continue to use it as the baseline against which C4.5's and CART's improvements are explained.

C4.5's gain ratio criterion successfully reduces the high-cardinality bias of information gain, but the reviewed literature notes that gain ratio can, in some circumstances, overcorrect and favour attributes with unusually low intrinsic information, a subtlety less frequently discussed than the original bias problem it was designed to solve. C4.5's post-pruning also depends on a pessimistic error estimate whose statistical assumptions are not always clearly validated against the properties of a given dataset.

CART's binary-split constraint is presented in the literature as both a strength (uniform treatment of categorical and continuous attributes, straightforward extension to regression) and a limitation (potentially deeper, less immediately readable trees on categorical data with several values, since a k -valued categorical attribute must be represented through multiple binary splits rather than one k -way split). CART's reliance on cross-validation for selecting the complexity parameter is computationally more expensive than C4.5's single-pass pruning, a trade-off that is acknowledged in efficiency-focused comparative studies but rarely quantified precisely across datasets of different sizes.

A broader critical observation is that most of the comparative studies reviewed rely on a relatively small number of benchmark datasets (frequently the same handful of UCI repository datasets), which limits the external validity of accuracy claims such as “CART outperforms ID3” or “C4.5 achieves the best accuracy.” Results that hold on small, clean benchmark datasets do not necessarily generalise to large-scale, high-dimensional, or highly imbalanced real-world data.

6. Research Gap

Synthesising the literature review and critical analysis above, three concrete gaps emerge:

- Lack of a unified, adaptive splitting criterion: existing comparative work treats information gain, gain ratio, and Gini impurity as fixed, mutually exclusive choices tied to a specific algorithm, rather than exploring criteria that could adapt per node based on the local distribution of attribute types (categorical vs. continuous), cardinality, and noise level.
- Limited benchmark diversity: most reviewed comparative accuracy claims are drawn from a small set of frequently reused benchmark datasets, leaving open questions about how the three algorithms behave on large-scale, high-dimensional, or class-imbalanced real-world datasets typical of contemporary applications such as healthcare monitoring or cybersecurity.
- Underexplored interaction between pruning strategy and interpretability: while accuracy and overfitting are frequently discussed, few reviewed studies quantify how differently each pruning strategy affects the resulting tree's size and hence its interpretability — a factor that matters greatly in regulated domains where model transparency is a requirement rather than a preference.

These gaps collectively point to the absence, in the reviewed literature, of a single adaptive framework that jointly optimises for accuracy, robustness to noise, and interpretability, rather than treating these as separate, independently studied properties of three fixed historical algorithms.

7. Proposed Research Direction

Based on the identified gap, this review proposes a hybrid, validation-aware attribute-selection and pruning framework, outlined here as a feasible direction for future research rather than a fully developed algorithm:

6. Adaptive per-node splitting criterion: at each node, evaluate candidate splits using both gain ratio (favoured for high-cardinality categorical attributes) and Gini impurity

(favoured for continuous or binary attributes), and select whichever criterion yields the larger statistically validated impurity reduction on a held-out validation fold, rather than fixing one criterion for the entire tree.

7. Statistically grounded, data-size-aware stopping rule: replace fixed minimum-sample or fixed-depth stopping thresholds with a rule that scales with the local sample size at each node (e.g., a chi-squared or permutation-test-based significance check), reducing the risk of both premature stopping on small nodes and unnecessary growth on large, noisy nodes.
8. Cross-validated cost-complexity pruning generalised to multi-way trees: extend CART style cost-complexity pruning, currently defined for strictly binary trees, to the multi-way tree structures produced by information-gain/gain-ratio splits, so that C4.5-style trees can benefit from the same rigorously validated pruning procedure that currently gives CART its robustness advantage.
9. Benchmark expansion: evaluate the resulting hybrid tree against ID3, C4.5, and CART on a broader and more diverse benchmark suite that deliberately includes large-scale, high-dimensional, and class-imbalanced datasets, to test whether accuracy and robustness advantages observed on classical benchmarks generalise.
10. Interpretability metrics alongside accuracy: report tree size (nodes/leaves) and average decision-path length as first-class evaluation metrics alongside accuracy, precision, recall, and F1-score, so that any accuracy gain from the hybrid criterion can be explicitly weighed against any loss of interpretability.

This direction is deliberately incremental and implementable: it combines mechanisms already validated independently in the reviewed literature (gain ratio, Gini impurity, cost-complexity pruning, cross-validation) rather than proposing an entirely new, unvalidated learning paradigm, which keeps the proposal grounded in demonstrated prior results while still addressing the specific gap identified in Section 6.

8. Discussion

The comparative and critical analysis above suggests that the choice among ID3, C4.5, and CART is rarely a question of one algorithm being unconditionally superior; rather, it is a question of which combination of attribute-selection criterion and pruning strategy best matches the statistical characteristics of a given dataset. For clean, purely categorical datasets where interpretability and training speed are the priority, ID3 remains a reasonable pedagogical and practical baseline. For general-purpose classification tasks involving a mix of categorical and continuous attributes, C4.5's gain ratio and post-pruning provide a good balance of accuracy and readability. For tasks requiring both classification and regression, or

where robustness to overfitting through rigorous cross-validation is paramount — including as a base learner inside bagging and boosting ensembles — CART's binary-split, cost-complexity pruned design is generally preferred, which is consistent with its widespread adoption inside Random Forest and Gradient Boosting implementations.

These findings carry broader implications for AI system design. As machine learning is increasingly deployed in regulated domains (healthcare, finance, criminal justice, and public administration), the demand for models that are simultaneously accurate and explainable continues to grow. The classical decision tree algorithms reviewed here demonstrate that interpretability need not be sacrificed for accuracy if pruning and attribute-selection strategies are chosen carefully — a lesson that remains directly relevant to current explainable-AI (XAI) research, much of which seeks to approximate complex black-box models using decision-tree like surrogate structures precisely because of the interpretability properties analysed in this review.

The proposed hybrid research direction in Section 7 is intended to extend, rather than discard, this classical foundation: by allowing the splitting criterion and pruning procedure to adapt to local data characteristics while still reporting interpretability metrics alongside accuracy, future work can make more defensible, evidence-based recommendations about which decision-tree configuration best serves a given application, instead of relying on the generalised, benchmark

limited claims identified as a gap in the current literature.

9. Conclusion

This review examined ID3, C4.5, and CART — the three classical algorithms that continue to underpin modern decision tree learning and its ensemble descendants. The comparison across attribute-selection method, pruning strategy, handling of continuous and missing values, interpretability, computational complexity, and reported classification performance shows a clear historical progression: ID3 established the information-gain paradigm but lacks pruning and continuous-attribute support; C4.5 corrected these limitations through gain ratio, threshold based continuous splitting, and post-pruning; and CART generalised the approach further to binary splits, regression support, and cross-validated cost-complexity pruning. The critical analysis identified that all three remain greedy, locally optimal procedures, and that most comparative accuracy claims in the literature rest on a narrow set of benchmark datasets. The identified research gap — the absence of a unified, adaptive, and benchmark-diverse framework that jointly optimises accuracy, robustness, and interpretability — motivates the proposed hybrid, validation-aware splitting and pruning framework outlined in this review. Future empirical work implementing and testing this proposal would help clarify whether

classical decision tree strategies can be meaningfully unified without sacrificing the interpretability that has made them enduringly relevant across more than three decades of AI research.

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Evaluation Rubrics:

Criterion	Weightage (%)	Excellent
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies research problem, objectives, scope, and relevance of learning approach.

2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.

4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
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5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters .	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.

7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
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8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.